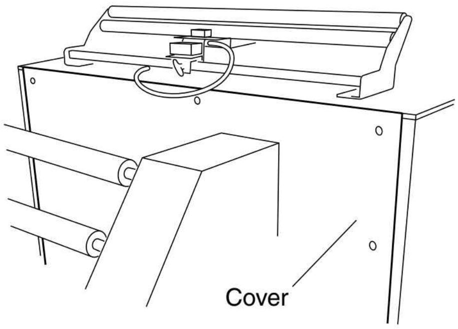
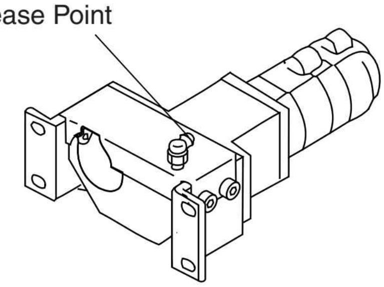
5. Open the rear upper cover by removing ten screws.



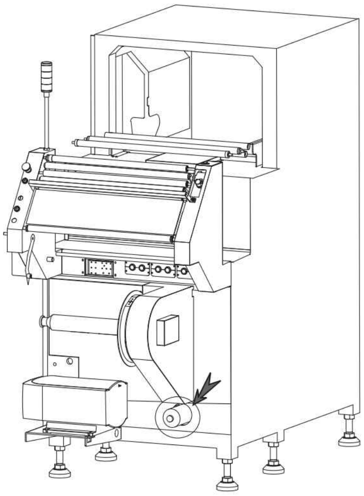
6. Grease the outer gears.



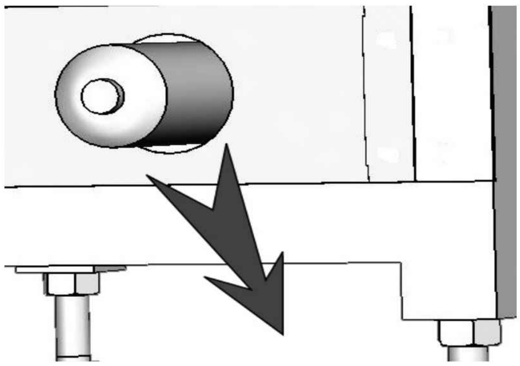
3.3 6-Month Inspections

3.3.1 Vacuum System Air Filter

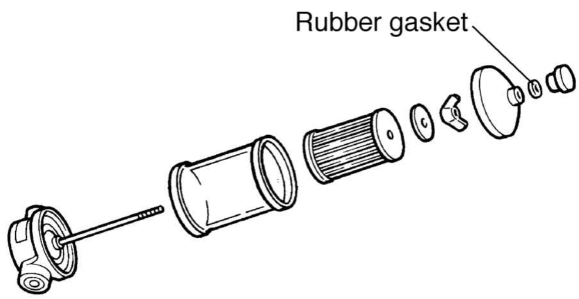
1. Turn the main power OFF.



2. Remove the air filter.
3. Using an air gun, blow out anything strange that may be adhering to the air filter.



4. Attach the air filter.
5. Attach the outer cover.



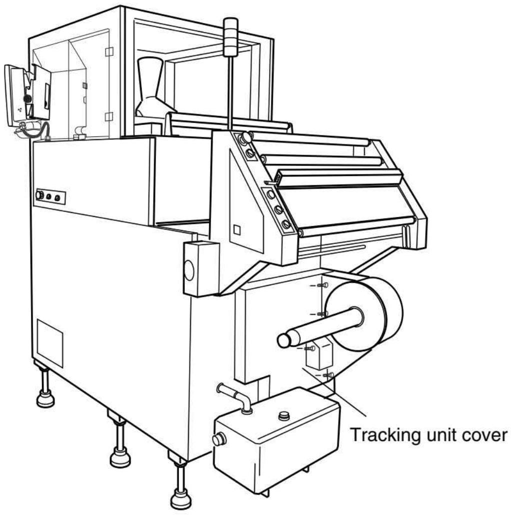
3.3.2 Lubrication

CAUTION

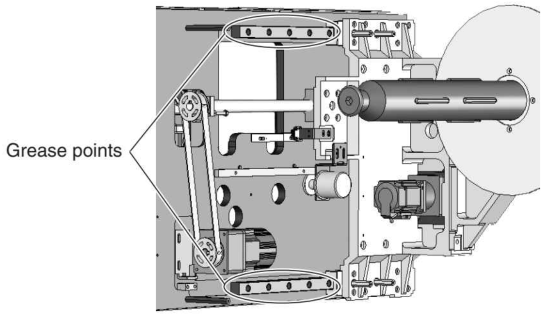
- **Lubricate periodically.**

3.3.2.1 Film Tracking Linear Guides

1. Turn the main power OFF.
2. Remove the film tracking unit cover.



3. Grease the film tracking unit linear guides.



4 REPLACEMENT AND ADJUSTMENT

4 Replacement and Adjustment

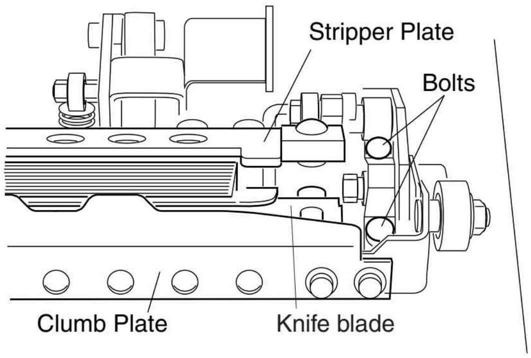
DANGER

- **Before opening the control panel cover, turn the main power OFF. Failure to do so can result in electrical shock.**

4.1 End Seal Unit

4.1.1 Replacing the Jaw Faces

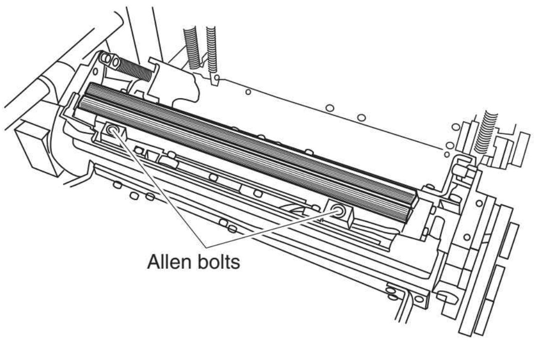
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.
5. Remove the allen bolts and remove the knife blade.



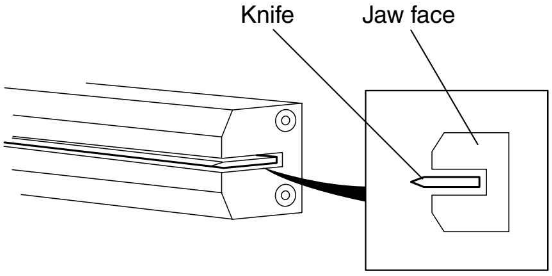
WARNING

- **Be careful when removing the knife blade. It is very sharp and hot.**

6. Replace the front and rear jaw faces with new ones.



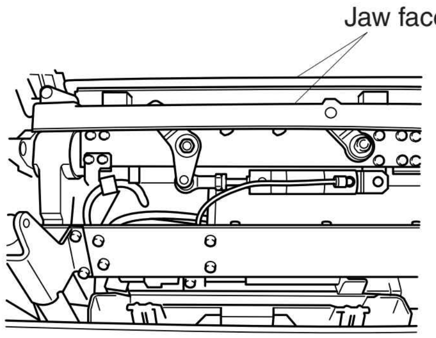
- Install the front jaw face so that the knife blade is in the middle of its knife slot.
- Install the rear jaw face temporary.



7. Make sure that the front and rear jaw faces are properly aligned.

   If the alignment is improper, adjust the position of the jaw. (For details on how to adjust the jaw, see "4.1.2 Jaw Alignment")

8. Tighten the allen bolts of the rear jaw.



4.1.2 Jaw Alignment

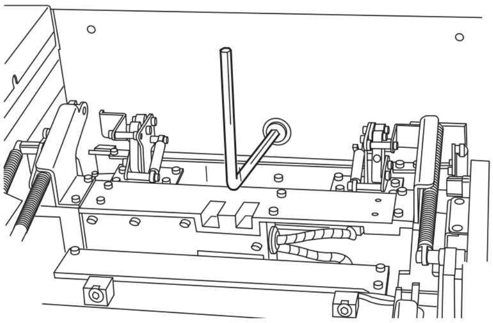
WARNING

- **It is sometimes necessary to have the main power ON when making adjustments. At such times, when the main power is ON, make sure that the motor power supply is turned OFF. If the motor power supply is left ON, the motor could start up, injuring someone. In addition, wear protective gloves when adjusting electrical equipment.**

1. Jog the jaw until the front and rear jaws engage.
2. If the following cases appear, jaw adjustment will be needed.
   a. Jaw alignment has upward or downward deviation.
   b. Jaw alignment has play.
   c. Jaw alignment has horizontal deviation.

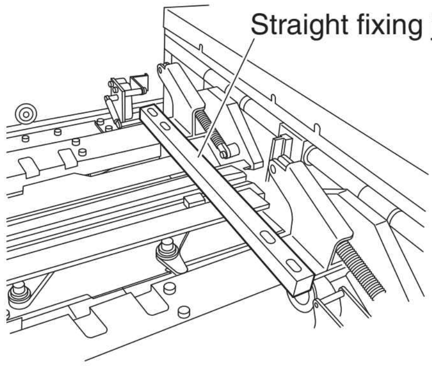
   To adjust the Jaw alignment, follow the below steps.

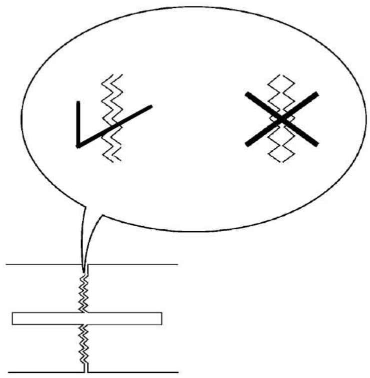
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.
5. Turn the pivot point screw to close jaw.



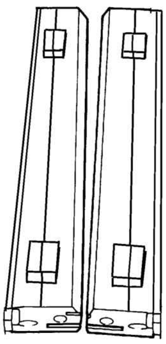
6. Adjust the front and rear jaw to match the knife.
7. Attach a straight fixing jig to keep jaw horizontal.
8. Check alignment with mirror.

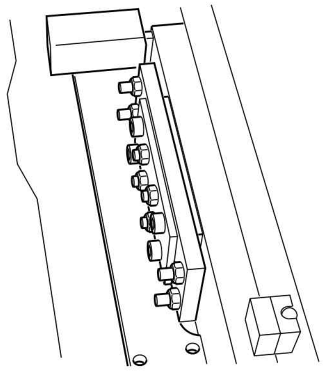
   If the jaw alignment is loose, adjust as follows.



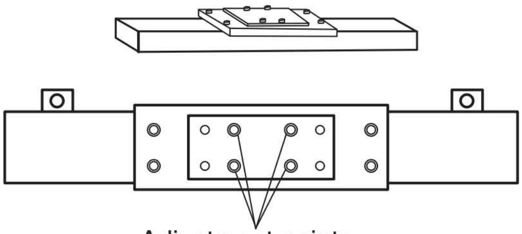


Right or Left adjustment

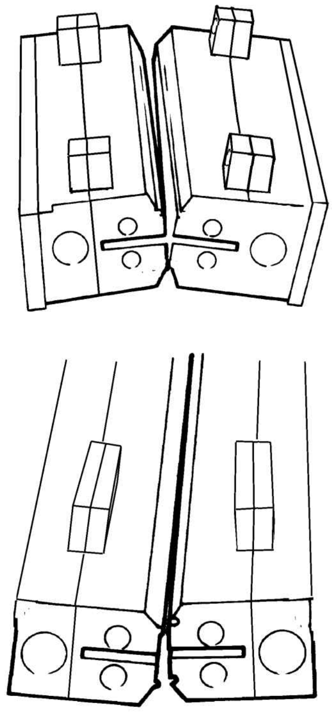




If the right and left are not aligned, adjust by turning adjustment screws on the metal plate attached to rear jaw.

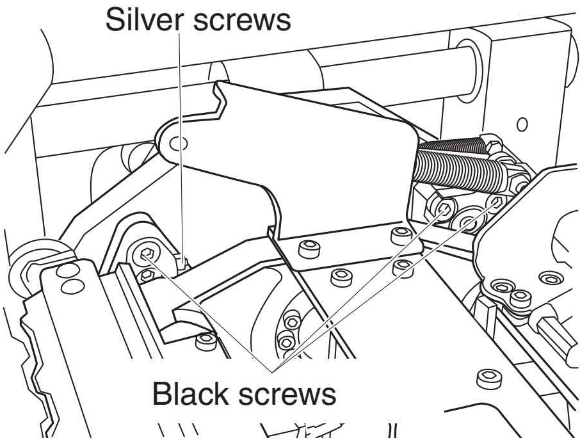


Top and Bottom adjustment



If top and bottom are not aligned, adjust the screw on the arm.

1. Loosen the 3 black screws.
2. Tighten or loosen the silver screw to adjust jaw face.



NOTE

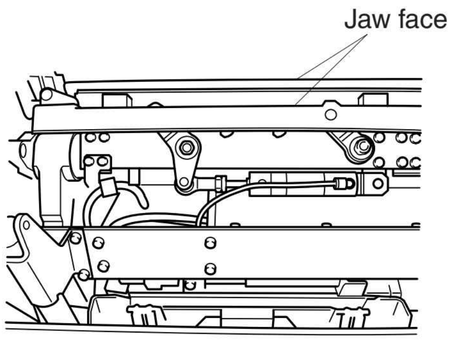
- There are two sets of jaw, so check both of jaw alignment.

4.1.3 M2 Motor

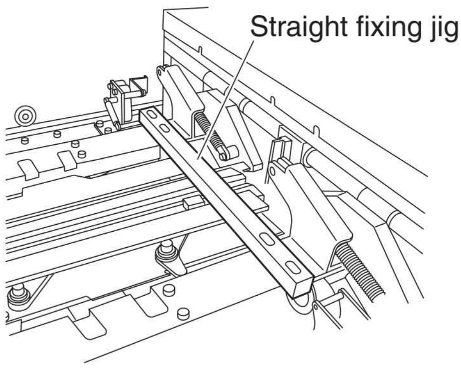
4.1.3.1 Timing Belt of M2 Motor

1. Proceed steps 1 to 4 of the section "4.1.1 Replacing the Jaw Faces".

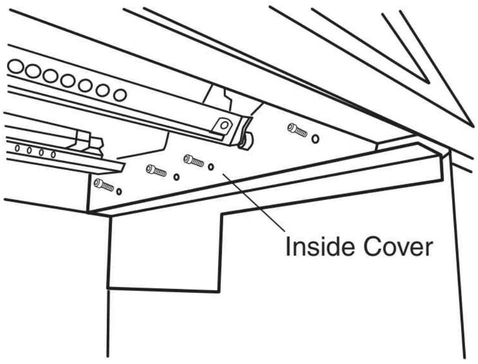
2. Using your hand, rotate the jaw arm and engage the front and rear jaws.



3. Attach a straight fixing jig to the jaw arm.



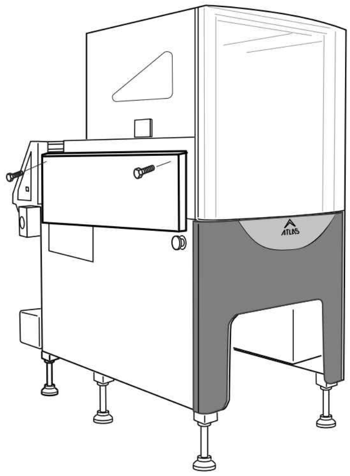
4. Remove 4 screws, remove the inside cover.



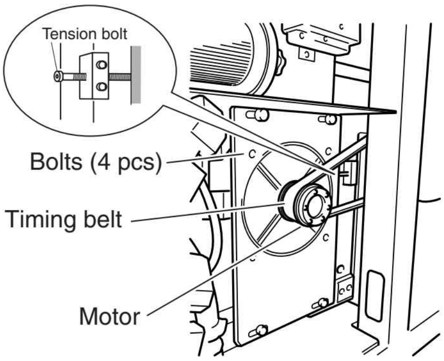
5. Remove the LEFT side cover.

CAUTION

- **Two people should remove the cover, since it is heavy.**

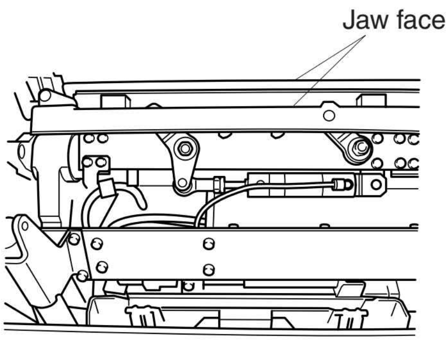


6. Loosen 4 bolts tightening the motor bracket.
7. Loosen the tension bolt.
8. Remove the timing belt.
9. Install the new timing belt.
10. Using your finger, press down on the center of the belt between pulleys. Adjust until the deflection is approximately 2 millimeters then tighten the tension bolt and tighten the motor bracket screws.
11. Attach the inside cover.

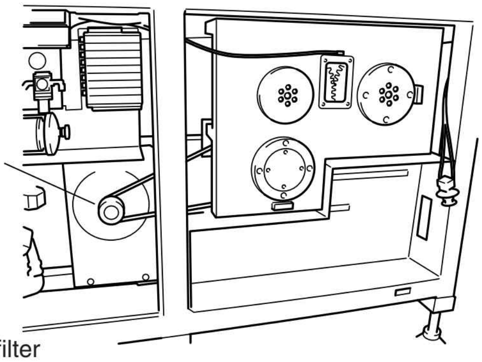


4.1.3.2 Replacing the M2 Motor

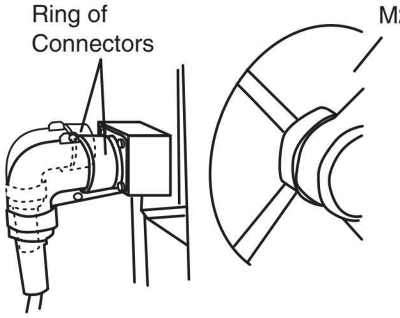
1. Proceed steps 1 to 4 of the section "4.1.1 Replacing the Jaw Faces".
2. Using your hand, rotate the jaw arm and engage the front and rear jaws.
3. Attach a straight fixing jig to the jaw arm.



4. Remove 4 screws and remove the noise filter unit so that you can access the M2 motor.



5. Loosen the ring, remove two connectors at the right of M2 motor.
6. Loosen 4 bolts tightened the motor bracket.
7. Loosen the tension bolt.
8. Remove 4 screws, remove the inside cover.
9. Remove the timing belt.
10. Loosen the mechanical lock, and remove the pulley from the M2 motor.
11. Remove the allen bolts, and remove the M2 motor.



CAUTION

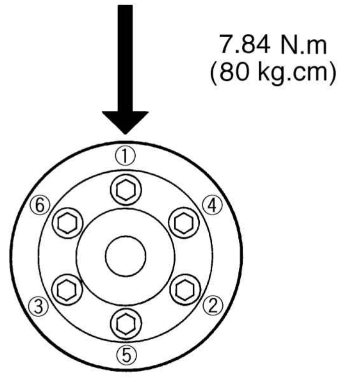
- **The motor is heavy and should be lifted by two people.**

12. Remove the motor from the motor base.
13. Attach the motor to the new motor base.

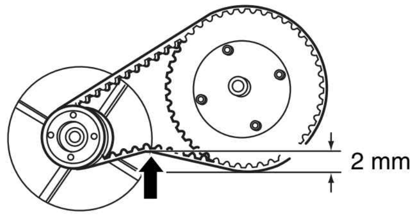
CAUTION

- **Do not cause any shock to the motor, or the motor's encoder can be damaged.**

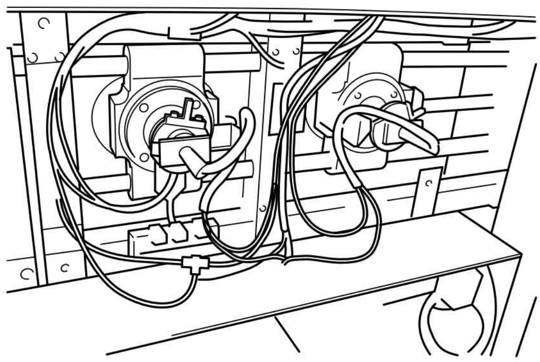
14. Attach and fasten the gear and mechanical lock to the motor shaft. Tighten the bolts for the mechanical lock at 7.84 N.m (80 kgf.cm) in a crisscross (star) pattern.
15. Install the motor with motor base.
16. Install the electrical box of the PS-0 unit.



17. Attach the timing belt.
18. Using your finger, press down on the center of the belt between gears. Adjust until the deflection is approximately 2 millimeters; then fasten the M2 motor.
19. Tighten the tension bolt.
20. Attach the inside cover.



4.1.3.3 Home Position of Jaw Rotation

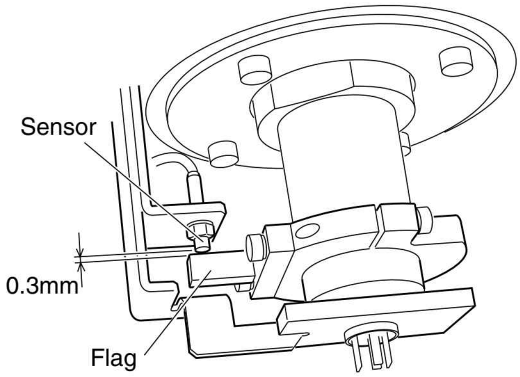


Home Position is the position where the jaw stops after the motor is turned on.
Then, either knife 1 or knife 2 comes upper side.

- When the rear sensor turned on, knife 1 comes to upper side.
- When the front sensor turned on, knife 2 comes to upper side.
- When Home position, the plate of the jaw should be angled 90 degree against the horizontal frame of main unit.

1. Turn the motor power supply OFF.
2. Turn the main power supply OFF.

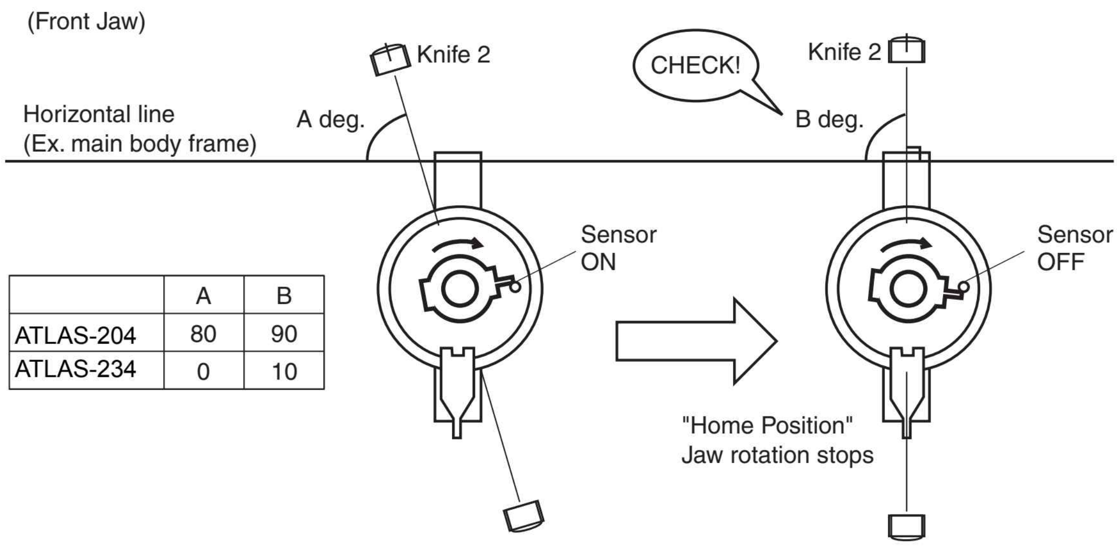
3. Open the left lower cover.
4. Set gap to 0.3mm between sensor and flag.



5. Set the angle of the jaw plate to A degree against the horizontal frame of main unit when the sensor turns on.
6. Turns the motor power on.

   Jaw rotates until the sensor switches on, then the jaw stops.

7. Confirm if the angle of the jaw plate is B degree against horizontal frame.



	A	B
ATLAS-204	80	90
ATLAS-234	0	10

8. If the angle is not B degree, adjust the flag, then restart the motor power on, and confirm the angle when the jaw stops at home position.

NOTE

- Repeat the above step until the angle to be correct.

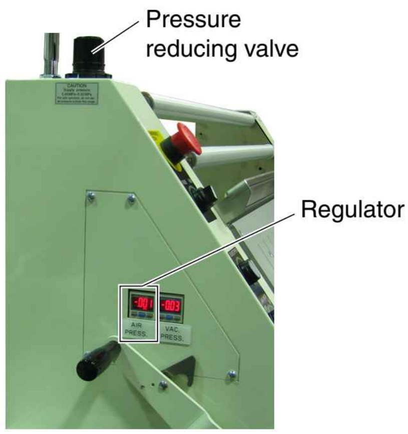
9. Adjust both of the Knife 1 and Knife 2 jaw sensor.

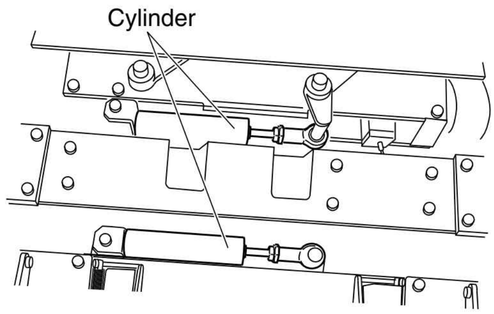
<How to adjust the flag>

1. Loosen the screws tightening the flag, and rotate the flag to appropriate angle.
2. Tighten the screws.

4.1.4 Replacing the Air Cylinder for Knife

1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.
5. Pull the knob for the pressure reducing valve and adjust until the pressure is 0.0 MPa.
6. Remove the rod end ; then replace the cylinder for knife with a new one.

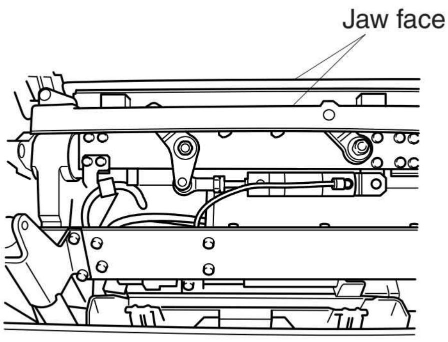




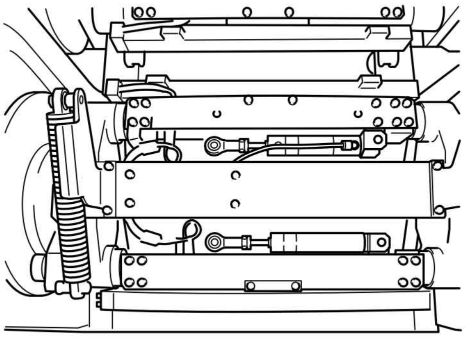
4.1.5 Jaw Heater

4.1.5.1 Replacing the Jaw Heater

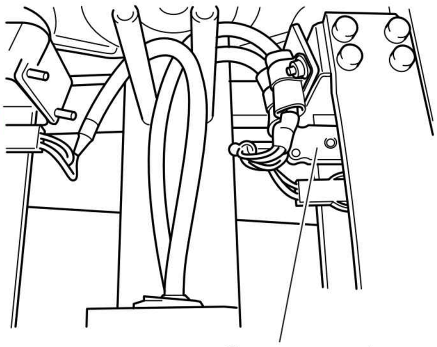
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.



5. Remove the sensor connector of the end seal heater from the plate.



6. Disconnect the cable that runs from the terminal connector for the heater.



CAUTION

- **The thermocouple has polarity and is identified by the color of the cable (red or white).
If a connection is made with the wrong polarity, the temperature control can fail or unit can be damage.**